Leeward Community College

Math and Science Division
Mechatronics Program

Course Syllabus

Course:	MECH151
CRN:	81143
Title:	Technical Design - Prototyping

Course Description

This course introduces the student to 3D modeling using CAD/CAM mechanical design automation software. Topics included are: sketching, orthographic projection, descriptive geometry, dimensioning, section views, auxiliary views, primary and secondary views, threads, fasteners, and production drawings. Students will build parametric models of parts and assemblies and make drawings of those parts and assemblies. The student will study CAD/CAM software configurations and translate parametric models to produce prototypes using various manufacturing methods. The course will also cover basic machine safety and operation.

Course Dates and Time

Credits:	4
Semester:	Fall 2026
Start date:	August 24, 2026
End date:	December 18, 2026
Time:	Wednesdays 1600-1830 (4:00 PM to 6:30 PM) Thursdays 1600-1850 (4:00 PM to 6:50 PM)
Classroom:	CE405
Office Hours:	By Appointment Only
Prerequisites:	None

Contact Information	**Instructor**	**Program Coordinator**	**Division**
Name:	Kainani Santos	William "Bill" Labby	N/A
Office location:	CE401	CE401	BS 106A
Email:	kainanis@hawaii.edu	wlabby@hawaii.edu	N/A
Phone number:	(808)600-8362	(808)330-2175	(808)455-0251

Course Learning Objectives

1. Demonstrate basic part modeling using 3D CAD software.
2. Demonstrate basic assembly modeling using 3D CAD software.
3. Simulate a complete assembly model using components.
4. Produce a component utilizing CAM software.
5. Produce a prototype from multiple substrates.

Course Information

Textbook/Reading Resources

Instructor will provide all materials in a digital format (i.e., PDF, video links). Students are responsible for making their own accommodations to modify the format of the content as needed (i.e., creating hardcopy printout of PDF).

Technology Requirements

Students must have reliable access to communication tools such as their university-provided Gmail account, the learning management system, Lamaku, and access to the internet outside of the designated course meeting times. Students should also plan to bring their own resources such as a laptop capable of running the course's CAD/CAM software. Issues with accessibility will be handled on a case-by-case matter.

Lecture, Lab, and Demonstrations

This course combines lecture instruction with hands-on CAD/CAM lab work and machine operation. Lab activities require that all participants wear proper PPE as designated by the instructor. Failure to have the proper PPE during machine use will result in the participant being asked to leave the class. Accommodations will be handled on a case-by-case matter and should be requested from the student to the instructor with as much notice as possible.

Grading Policy

Grades for the course will be calculated using the assessment rubric provided in this syllabus.

Assignment	**Total Per Course**	**Value per Assignment**	**Contribution**
Attendance	12	1	10%
Design Portfolio	1	100	50%
Unit Quizzes	4	5	10%
Prototype (Final Exam)	1	100	30%
Total			100%

While grades are earned individually, students of the course are encouraged to have open discussions with each other to build their own knowledge base on executing technical designs confidently.

Attendance via Practice Problem Submissions

These weekly practice problems are designed to encourage further hands-on practice and build competency in the CAD program you've chosen for this course. Attendance credit (12 total, 10% of course grade) is earned weekly by completing and submitting SolidWorks practice-problem work — see the course website's SolidWorks Practice Problems page for the full Level 1–8 problem sets and the Level 9 project handout. Email that week's deliverable to kainanis@hawaii.edu — as a direct attachment, or through the UH FileDrop service (hawaii.edu/filedrop) addressed to the same email — by the Sunday deadline below to receive that week's attendance credit.

Credit is contingent on accuracy: your model must actually capture the part's form — its features, proportions, and geometry as shown in the problem — not a placeholder or primitive shape (e.g. a plain block or cylinder) standing in for it. Submissions that don't reflect the modeled part will not receive credit.

Deadlines for Receiving Attendance Credit

Week	Dates	Practice Problem Focus	Deliverable Due	Deadline
4	Sep 16–17	Level 1 – Basic Sketch and Extrusion	3 problems of your choice from Level 1	Sun, Sep 20
5	Sep 23–24	Level 2 – Sketch Tools & End Conditions	3 problems from Level 2	Sun, Sep 27
6	Sep 30–Oct 1	Level 3 – Global Variables & Sketch Patterns	3 problems from Level 3	Sun, Oct 4
7	Oct 7–8	Level 4 – Extrude Cut & Fillet/Chamfer	3 problems from Level 4	Sun, Oct 11
8	Oct 14–15	Level 5 – Reference Geometry	3 problems from Level 5	Sun, Oct 18
9	Oct 21–22	Level 6 – Revolve Boss/Cut	3 problems from Level 6	Sun, Oct 25
10	Oct 28–29	Level 7 – Feature Patterning	3 problems from Level 7	Sun, Nov 1
11	Nov 4–5	Level 8 – Sweep Boss/Cut	3 problems from Level 8	Sun, Nov 8
12	Nov 11–12	Level 9 Vice – Stage 1	Mechanism concept finalized; fixed jaw/base modeled	Sun, Nov 15
13	Nov 18–19	Level 9 Vice – Stage 2	Moving jaw + actuator parts modeled	Sun, Nov 22
14	Nov 25–26*	Level 9 Vice – Stage 3	Full assembly with all parts mated	Sun, Nov 29
15	Dec 2–3	Level 9 Vice – Stage 4 (Final)	Full range of motion verified; final submission	Sun, Dec 6

*No class Thursday, November 26, 2026 (Thanksgiving Holiday).

Levels 1–8 and the Level 9 project follow the same schedule described in the Course Schedule below; this table exists to tie weekly attendance credit directly to practice-problem progress. This is a tentative pacing guide and may be adjusted by the instructor during the semester.

Final Exam Information

The final exam will be evaluated in the form of a tangible prototype wholly made by MECH151 students. As part of the final exam, groups may be formed so long as they range from 1-3 student members and teams must present their design. The following elements are part of the evaluation:

- Prototype Design Proposal – Establishing team members, what design is being made, and how it will:
    - Integrate of at least 2 different types of bulk material (i.e., thermoplastic + raw/processed stock).
    - Demonstrate of at least 1 joining method between 2 parts (i.e., heat insert, snap fit).
- All pertinent documentation for manufacturing (i.e., drawings with pertinent views and GD&T callouts).
- Work breakdown summary.
- Self Reflection.

The Final Exam will be held at CE405 on Monday, December 14, 2026, wherein groups will present their design to the class. The exam will begin at 1645 (4:45pm) and end at 1845 (6:45pm). Exam's must be submitted to the instructor by 1845 (6:45pm) to be considered for grading. Failure to submit the exam by the ending time will result in an automatic score of zero (0).

Grade Scale

Final letter grades are assigned based on the standard A-F grading scale.

Acceptable Use of Artificial Intelligence for Coursework

While the use of Artificial Intelligence, such as ChatGPT, can offer students guidance, it is essential to uphold academic integrity and allow students to be evaluated based on their own understanding of course topics. Furthermore, students are expected to meet all course learning objectives and adhere to the parameters outlined within this syllabus.

Usage of grammar and spell checking tools (i.e., Grammarly) or those integrated into processing software (i.e., Microsoft Word, Google Docs) may be used to enhance the communication of ideas by student.

Additional Notifications for Students

Academic Dishonesty

Academic dishonesty cannot be condoned by the University. Such dishonesty includes cheating and plagiarism (examples of which are given below), which violate the Student Conduct Code and could result in expulsion from the University.

Cheating includes but is not limited to giving unauthorized help during an examination, obtaining unauthorized information about an examination before it is administered, using inappropriate sources of information during an examination, altering the record of any grades, altering answers after an examination has been submitted, falsifying any official University record, and misrepresenting the facts in order to obtain exemptions from course requirements.

Plagiarism includes but is not limited to submitting any document, to satisfy an academic requirement, that has been copied in whole or part from another individual's work without identifying that individual; neglecting to identify as a quotation a documented idea that has not been assimilated into the student's language and style, or paraphrasing a passage so closely that the reader is misled as to the source; submitting the same written or oral material in more than one course without obtaining authorization from the instructors involved; or dry-labbing, which includes (a) obtaining and using experimental data from other students without the express consent of the instructor, (b) utilizing experimental data and laboratory write-ups from other sections of the course or from previous terms during which the course was conducted, and (c) fabricating data to fit the expected results.

Please see the Leeward CC website for more information about academic dishonesty.

Participation Verification

Students who do not participate in their classes during the first week of the semester are considered to be "No Shows." The College assumes that "No Show" students no longer wish to participate in their education. Students who are identified as "No Shows" will be administratively disenrolled from their classes and given a 100% tuition refund. Participant Verification helps to release students who registered but did not intend to come to class from a financial obligation and failing grade. It also keeps the College in compliance with federal financial aid guidelines.

See Executive Policy 7.209 Student Participation Verification in Coursework.

Title IX

Leeward Community College is committed to supporting students and upholding the College's non-discrimination policy. Under Title IX, discrimination based upon sex and gender is prohibited. If you experience an incident of sex- or gender-based discrimination, harassment, or violence, we encourage you to reach out for help. While you may talk to a faculty member, understand that as a "Responsible Employee" of the College, the faculty member must inform the college's Title IX Coordinator to ensure that our students are supported and aware of the resources available. If you would like to speak with someone who can afford you with privacy and confidentiality, there are designated confidential resources available who can meet with you.

For more information about available resources, and University policies, please see our Title IX website or contact our Title IX Coordinator, Thomas Hirsbrunner at 808-455-0478 or email leetix@hawaii.edu.

Course Schedule*

This schedule covers Weeks 4–16 (the instructor's period of coverage) and is organized into four units: CAD Fundamentals (Basic), CAD Advanced, Design for Assembly (DFA), and Design for Manufacturing (DFM).

Week	Date	Session	Topics/Assignments
Unit 1 – CAD Fundamentals (Basic)			
4	September 16, 2026	Wed	Sketching fundamentals and sketch constraints
4	September 17, 2026	Thu	Orthographic projection and multiview theory
5	September 23, 2026	Wed	Descriptive geometry basics
5	September 24, 2026	Thu	Primary and secondary views
6	September 30, 2026	Wed	Basic dimensioning standards
6	October 1, 2026	Thu	Introduction to parametric part modeling
7	October 7, 2026	Wed	Unit 1 Quiz
Unit 2 – CAD Advanced			
7	October 8, 2026	Thu	Section views (full, half, offset), Auxiliary views, Threads and fastener representation/callouts
8	October 14, 2026	Wed	
8	October 15, 2026	Thu	
9	October 21, 2026	Wed	Production/detail drawings (title blocks, BOM basics)
9	October 22, 2026	Thu	Introduction to tolerancing
10	October 28, 2026	Wed	CAD software configuration and CAD/CAM data translation
10	October 29, 2026	Thu	Unit 2 Quiz; Prototype Groups Form
Unit 3 – Design for Assembly (DFA)			
11	November 4, 2026	Wed	Assembly modeling (mates/constraints), Building assemblies from part models, Applying fasteners and threads within assemblies
11	November 5, 2026	Thu	
12	November 11, 2026	Wed	DFA principles (part count reduction, ease of assembly, tolerance stack-up)
12	November 12, 2026	Thu	Assembly drawings, exploded views, and BOMs; Design Proposal
13	November 18, 2026	Wed	Design review/iteration workshop (based on submitted proposals)
13	November 19, 2026	Thu	Unit 3 Quiz
Unit 4 – Design for Manufacturing (DFM)			
14	November 25, 2026	Wed	Translating parametric models to manufacturing-ready files
14	November 26, 2026	Thu	NO CLASS – Thanksgiving Holiday
15	December 2, 2026	Wed	Manufacturing methods survey: additive, subtractive, DFM considerations, Machine safety, PPE, and operation
15	December 3, 2026	Thu	Prototype work time
16	December 9, 2026	Wed	Prototype work time
16	December 10, 2026	Thu	Unit 4 Quiz
Final Exam			
17	December 14, 2026	Monday	Final Exam

**Tentative schedule that may change during the semester. Updates will be provided by instructor.*